Engineering Probability
Homework 4

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October 1st 2025

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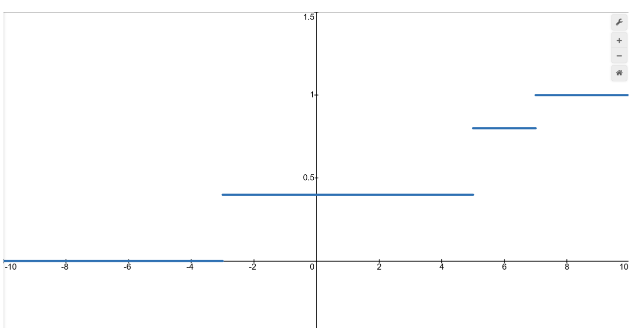
(1)

The random variable X has cumulative distribution function (CDF)

$$F_X(x) = \begin{cases} 0 & x < -3 \\ 0.4 & -3 \leq x < 5 \\ 0.8 & 5 \leq x < 7 \\ 1 & x \geq 7 \end{cases}$$

(a) Sketch the graph of the CDF.



(b) Determine $P_X(x)$, the probability mass function (PMF) of X.

$$P_X(x) = \begin{cases} 0.4 & x = -3, 5 \\ 0.2 & x = 7 \\ 0 & \text{otherwise} \end{cases}$$

(c) Find the expected value, $E[X]$ of the random variable X.

$$E[X] = \sum_{x \in S_X} x \cdot P_X(x) = -3(0.4) + 5(0.4) + 7(0.2) = 2.2$$

(d) Find the variance, $\text{Var}[X]$, of the random variable X.

$$\text{Var}[X] = E[X^2] - (E[X])^2 = \left((-3)^2(0.4) + (5)^2(0.4) + (7)^2(0.2)\right) - (2.2)^2 = 18.56$$

(2)

At the One Top Pizza shop, a pizza sold has mushrooms with probability $p = \frac{2}{3}$. On a day in which 100 pizzas are sold, let N be the number of pizzas sold before the first pizza with mushrooms is sold. What is the PMF of N? What is the CDF of N?

$$\text{PMF: } P_N(n) = \begin{cases} \left(\frac{1}{3}\right)^n \left(\frac{2}{3}\right) & n = 0, 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$\text{CDF: } F_N(n) = \begin{cases} 1 - \left(\frac{1}{3}\right)^{n+1} & n = 0, 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

(3)

Let X have the uniform PMF

$$P_X(x) = \begin{cases} 0.01 & x = 1, 2, \ldots, 100 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find a mode x_{mod} of X. If the mode is not unique, find the set X_{mod} of all modes of X.

The mode is not unique b/c it's a uniform PMF, $X_{\text{mod}} = \{1, 2, \ldots, 100\}$.

(b) Find a median x_{med} of X. If the median is not unique, find the set X_{med} of all numbers x that are medians of X.

$x_{med} = \{50, 51\}$

(4)

Voice calls cost 20 cents each and data calls cost 30 cents each. C is the cost of one telephone call. The probability that a call is a voice call is $P[V] = 0.6$. The probability of a data call is $P[D] = 0.4$.

(a) Determine $P_C(c)$, the PMF of C.

$$P_C(c) = \begin{cases} 0.6 & c = 20 \\ 0.4 & c = 30 \\ 0 & \text{otherwise} \end{cases}$$

(b) Determine $E[C]$, the expected value of C.

$$E[C] = 20(0.6) + 30(0.4) = 24$$

(c) Determine $\text{Var}[C]$, the variance of C.

$$\text{Var}[C] = E[(C - \mu_X)^2] = \sum_{c \in S_C} (c - \mu)^2 \cdot P_C(c) = (20 - 24)^2(0.6) + (30 - 24)^2(0.4) = 24$$

(5)

Given the random variable X in Problem 1, let $W = g(X) = -X$.

(a) Determine $P_W(w)$.

$$P_W(w) = \begin{cases} 0.4 & w = 3, -5 \\ 0.2 & w = -7 \\ 0 & \text{otherwise} \end{cases}$$

(b) Determine $F_W(w)$.

$$F_W(w) = \begin{cases} 0 & w < -7 \\ 0.2 & -7 \leq w < -5 \\ 0.6 & -5 \leq w < 3 \\ 1 & w \geq 3 \end{cases}$$

(c) Determine $E[W]$.

$$E[W] = \sum_{x \in S_X} -x \cdot P_X(x) = -E[X] = -2.2$$

(d) Determine $\mathrm{Var}[W]$.

$$\mathrm{Var}[W] = (-1)^2 \mathrm{Var}[X] = 18.56$$

(6)

At a discount brokerage, a stock purchase or sale worth less than \$10,000 incurs a brokerage fee of 1% of the value of the transaction. A transaction worth more than \$10,000 incurs a fee of \$100 plus 0.5% of the amount exceeding \$10,000. Note that for a fraction of a cent, the brokerage always charges the customer a full penny. You wish to buy 100 shares of a stock whose price D in dollars has a PMF

$$P_D(d) = \begin{cases} \frac{1}{3} & d = 99.75,\ 100,\ 100.25 \\ 0 & \text{otherwise} \end{cases}$$

What is the PMF of C, the cost of buying stock (including brokerage fee).

$$d = 99.75, 100, 100.25$$

$$c_{99.75} = 9,975 + (9975 \cdot 1\%) = 10,074.75$$

$$c_{100} = 10,000 + (10,000 \cdot 1\%) = 10,100$$

$$c_{100.25} = 10,025 + (100 + 25 \cdot 0.5\%) = 10,125.13 \text{ (rounded up)}$$

$$P_C(c) = \begin{cases} \frac{1}{3} & c = 10074.75,\ 10100,\ 10125.13 \\ 0 & \text{otherwise} \end{cases}$$

(7)

Let X be a discrete random variable with PMF

$$P_X(x) = \begin{cases} \frac{1}{8}, & x = 1, 2, 3, 4, \\ \frac{1}{2}, & x = 5, \\ 0, & \text{otherwise.} \end{cases}$$

Define a new random variable

$$Y = g(X) = \begin{cases} 0, & x \in \{1, 2\}, \\ 1, & x \in \{3, 4, 5\}. \end{cases}$$

(a) Determine the PMF of Y.

$$P_Y(y) = \begin{cases} \frac{1}{4}, & y = 0, \\ \frac{3}{4}, & y = 1, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Derive the CDF of Y.

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \frac{1}{4}, & 0 \le y < 1, \\ 1, & y \ge 1. \end{cases}$$

(c) Compute the expected value $E[Y]$.

$$E[Y] = 0\left(\frac{1}{4}\right) + 1\left(\frac{3}{4}\right) = \frac{3}{4}$$

(d) Determine the variance $\text{Var}[Y]$ and the standard deviation of Y.

$$\text{Var}[Y] = E[Y^2] - (E[Y])^2 = 0^2\left(\frac{1}{4}\right) + 1^2\left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 = \frac{3}{16}$$